

LAST NAME: _____

Practice Quiz 02D — MATH 2055 — Fall 2026

Practice for Quiz 02

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Auto-generated by the Codex harness using GPT-5.

1. Let R mean *the robot moves* and E mean *the emergency stop is off*. The statement is $R \rightarrow E$.

(a) Write its converse and contrapositive as formulas. Make one four-row truth table and say which is equivalent to $R \rightarrow E$.

- (b) In $\exists z \in \mathbf{Z} \forall w \in \mathbf{Z} (z + w = d)$, list the free variables and the bound variables.